

Math 306 Midterm 2

Name: _____

1. State the definitions of:

- a. An eigenvector for $T \in \mathcal{L}(V)$.
- b. A T -invariant subspace.
- c. An inner product on a vector space V over the field $\mathbb{C}$.
- d. A diagonalizable linear map $T \in \mathcal{L}(V)$.

2. Let $T \in \mathcal{L}(V)$. (Re)prove there is a basis for V such that the matrix for T is upper triangular.

3. Let $T \in \mathcal{L}(P_2(\mathbb{C}))$ be defined by $T(p(x)) = p(x) + p(-x)$. Find the eigenvalues/vectors for T . Is T diagonalizable?

4. Prove that if x, y are nonzero orthogonal vectors in a inner product vector space V , then x and y are linearly independent.

5. Proof or counterexample: $U = \{T \in \mathcal{L}(V, W) : T \text{ is an isomorphism}\}$ a subspace of $\mathcal{L}(V, W)$.